

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: CSA51

COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO5: *Develop the network security strategies based on the study of viruses, worms, firewall architectures, and IDS/IPS functionalities.CO (BL5 , BL6)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

TOTAL MARKS: 50

Annexure A

PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

Code of Conduct:

*I **Kartheeka Repalle(192372289)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:
Kartheeka Repalle

Annexure A

PSEUDOCODE WRITING TASK (ALGORITHM DESIGN)

1. Write pseudocode to simulate a simplified TLS handshake process, including client request, server authentication using a digital certificate, session key generation, and secure data exchange.
2. Develop pseudocode for a firewall rule engine that filters incoming packets based on IP address, port number, and protocol, and decides whether to allow or block traffic.
3. Write pseudocode to detect and classify malware behavior (virus, worm, or Trojan) based on propagation method, file modification patterns, and user interaction.
4. Design pseudocode for an Intrusion Detection System (IDS) that uses both signature-based and anomaly-based detection to identify suspicious network activity.
5. Create pseudocode for an Intrusion Prevention System (IPS) that monitors network traffic in real time and automatically blocks malicious requests while logging intrusion attempts.

1. SIMPLIFIED TLS HANDSHAKE – PSEUDOCODE

ANSWER:

BEGIN

Client sends connection request to Server

Server receives the request

Server sends its digital certificate to Client

Client verifies the server certificate

IF certificate is valid THEN

Client accepts the Server

ELSE

Terminate connection

END IF

Client and Server perform key exchange

Generate a shared session key

Server and Client confirm successful key establishment

Client encrypts application data using the session key

Client sends encrypted data to Server

Server decrypts the data using the session key

Server sends encrypted response to Client

Client decrypts the response

Secure communication continues

END

2. FIREWALL RULE ENGINE – PSEUDOCODE

ANSWER:

BEGIN

Receive incoming network packet

Read source IP address

Read destination IP address

Read destination port number

Read network protocol

FOR each firewall rule DO

 Compare packet IP address with rule

 Compare packet port number with rule

 Compare packet protocol with rule

 IF all required conditions match THEN

 IF rule action = ALLOW THEN

 Allow packet

 Log allowed traffic

 ELSE IF rule action = BLOCK THEN

 Block packet

 Log blocked traffic

 END IF

 Stop checking further rules

 END IF

END FOR

IF no rule matches THEN

 Apply default firewall policy

 IF default policy = ALLOW THEN

 Allow packet

 ELSE

 Block packet

 END IF

END IF

END

3. MALWARE BEHAVIOR CLASSIFICATION – PSEUDOCODE

ANSWER:

BEGIN

Collect malware behavior information

Check propagation method

Check file modification behavior

Check user interaction requirements

IF malware automatically spreads
through networks or connected systems THEN

Classify as WORM

ELSE IF malware attaches itself to legitimate
files and modifies or infects those files THEN

Classify as VIRUS

ELSE IF malware appears as or is delivered
through a legitimate-looking program
and requires user execution or interaction THEN

Classify as TROJAN

ELSE

Classify as UNKNOWN MALWARE

END IF

Display malware classification

Record observed behavior

END

4. INTRUSION DETECTION SYSTEM (IDS) – PSEUDOCODE

ANSWER:

BEGIN

Start IDS monitoring

Capture incoming and outgoing network traffic

FOR each network packet or network event DO

 Extract relevant traffic information

 Perform signature-based detection

 IF traffic matches a known attack signature THEN

 Generate HIGH-PRIORITY ALERT

 Record signature-based detection

 END IF

 Perform anomaly-based detection

 Compare traffic behavior with normal baseline

```
IF traffic significantly differs from normal behavior THEN
    Generate ANOMALY ALERT
    Record anomaly details
END IF
```

```
IF no suspicious activity is detected THEN
    Mark traffic as NORMAL
END IF
```

```
Store event in IDS log
```

```
END FOR
```

```
END
```

5. INTRUSION PREVENTION SYSTEM (IPS) – PSEUDOCODE

ANSWER:

```
BEGIN
```

```
Start real-time network monitoring
```

```
Capture incoming network request
```

```
Inspect packet and request information
```

```
Check request against known malicious signatures
```

```
Perform anomaly and behavior analysis
```

```
IF request is identified as malicious THEN
```

```
    Block the request
```

Record source information

Record reason for blocking

Generate security alert

ELSE IF request is suspicious THEN

 Apply configured security policy

 Log the suspicious activity

 IF policy requires blocking THEN

 Block request

 ELSE

 Allow request and continue monitoring

 END IF

ELSE

 Allow the request

END IF

Continue monitoring network traffic

END